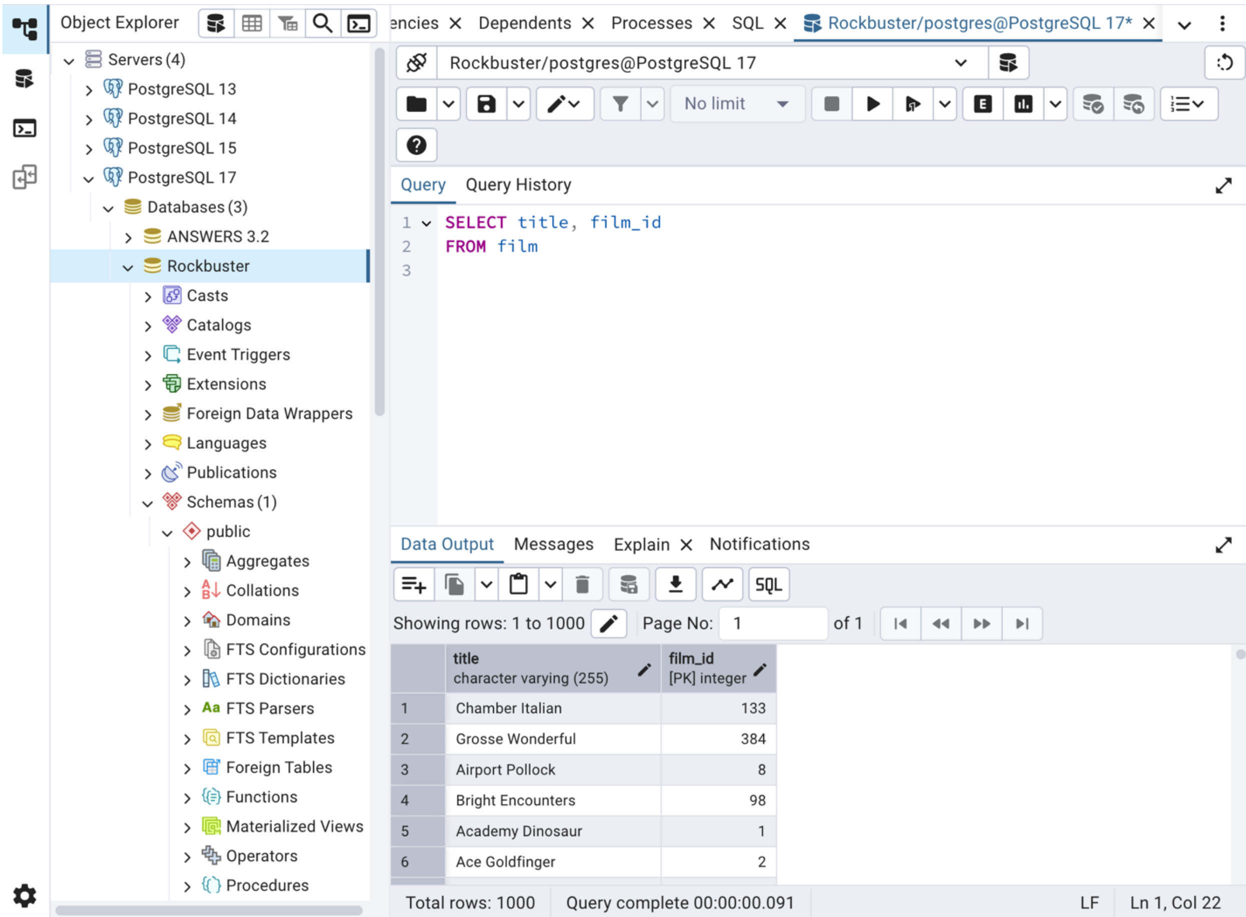
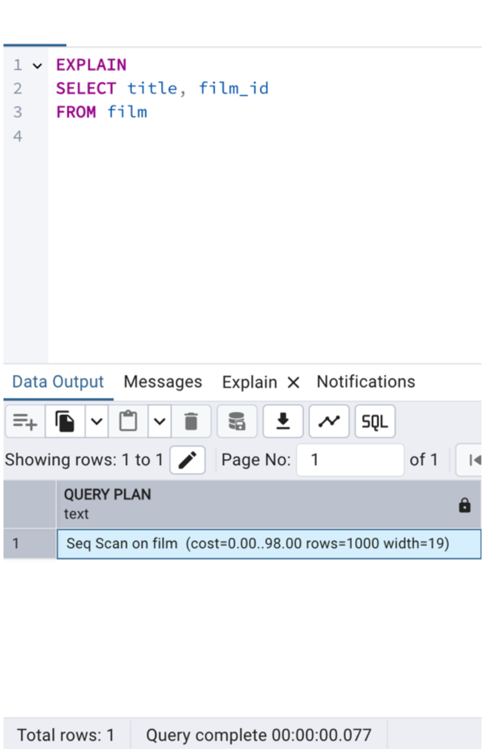
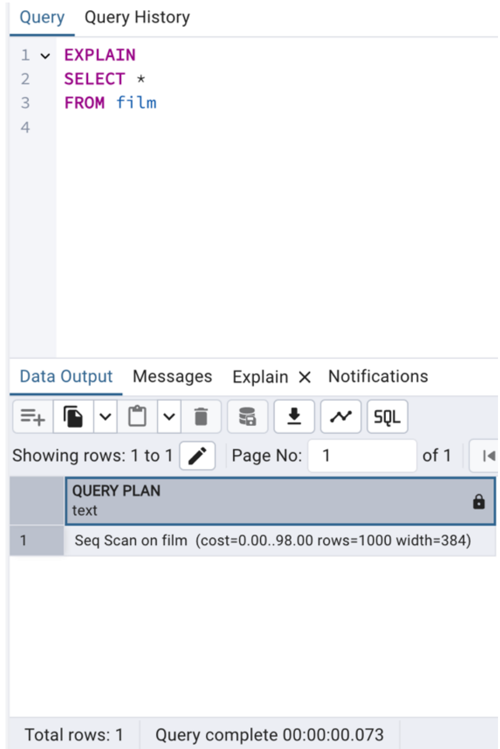
<u>Exercise 3.5</u> Yaw Assensoh Opoku

1a

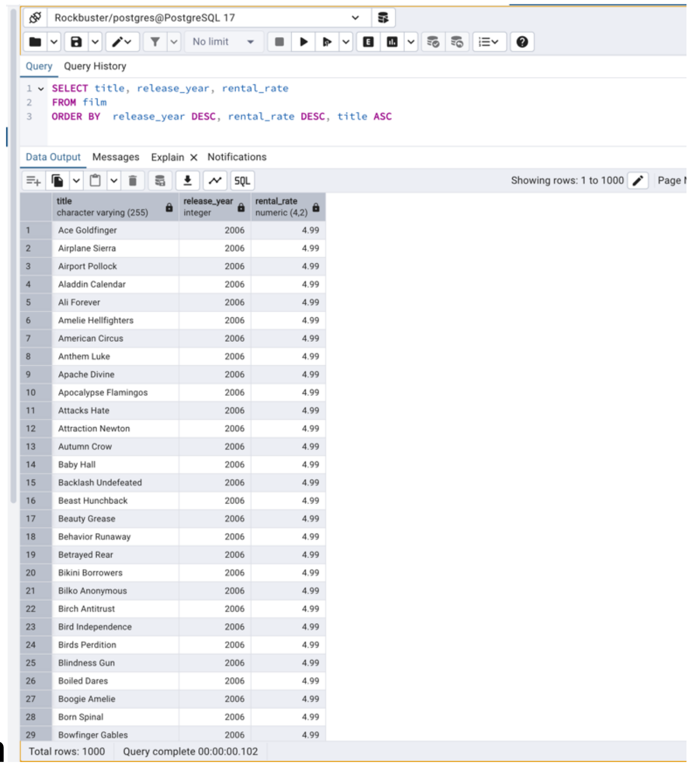


1b.





1c. These two queries are basically the same with costs and rows but different from their width but the original query retrieves all columns whiles the revised query retrieved from title and film_id columns. For optimization I'd suggest using LIMIT instead.



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No limit

Query | Query History

```sql
SELECT title, release_year, rental_rate
FROM film
ORDER BY  release_year DESC, rental_rate DESC, title ASC
```

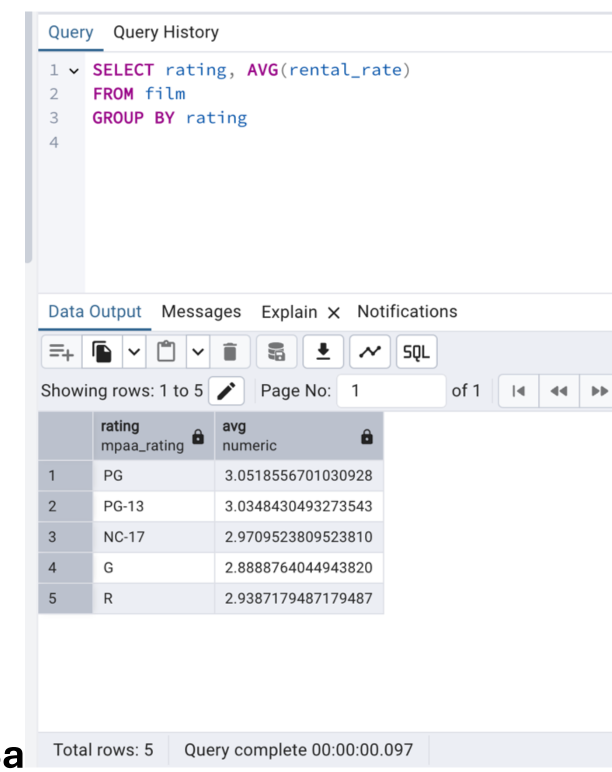
Data Output | Messages | Explain | Notifications

Showing rows: 1 to 1000

	title character varying (255)	release_year integer	rental_rate numeric (4,2)
1	Ace Goldfinger	2006	4.99
2	Airplane Sierra	2006	4.99
3	Airport Pollock	2006	4.99
4	Aladdin Calendar	2006	4.99
5	Ali Forever	2006	4.99
6	Amelie Hellfighters	2006	4.99
7	American Circus	2006	4.99
8	Anthem Luke	2006	4.99
9	Apache Divine	2006	4.99
10	Apocalypse Flamingos	2006	4.99
11	Attacks Hate	2006	4.99
12	Attraction Newton	2006	4.99
13	Autumn Crow	2006	4.99
14	Baby Hall	2006	4.99
15	Backlash Undefeated	2006	4.99
16	Beast Hunchback	2006	4.99
17	Beauty Grease	2006	4.99
18	Behavior Runaway	2006	4.99
19	Betrayed Rear	2006	4.99
20	Bikini Borrowers	2006	4.99
21	Bilko Anonymous	2006	4.99
22	Birch Antitrust	2006	4.99
23	Bird Independence	2006	4.99
24	Birds Perdition	2006	4.99
25	Blindness Gun	2006	4.99
26	Boiled Dares	2006	4.99
27	Boogie Amelie	2006	4.99
28	Born Spinal	2006	4.99
29	Bowfinger Gables	2006	4.99

Total rows: 1000 | Query complete 00:00:00.102

2a



Query | Query History

```sql
SELECT rating, AVG(rental_rate)
FROM film
GROUP BY rating
```

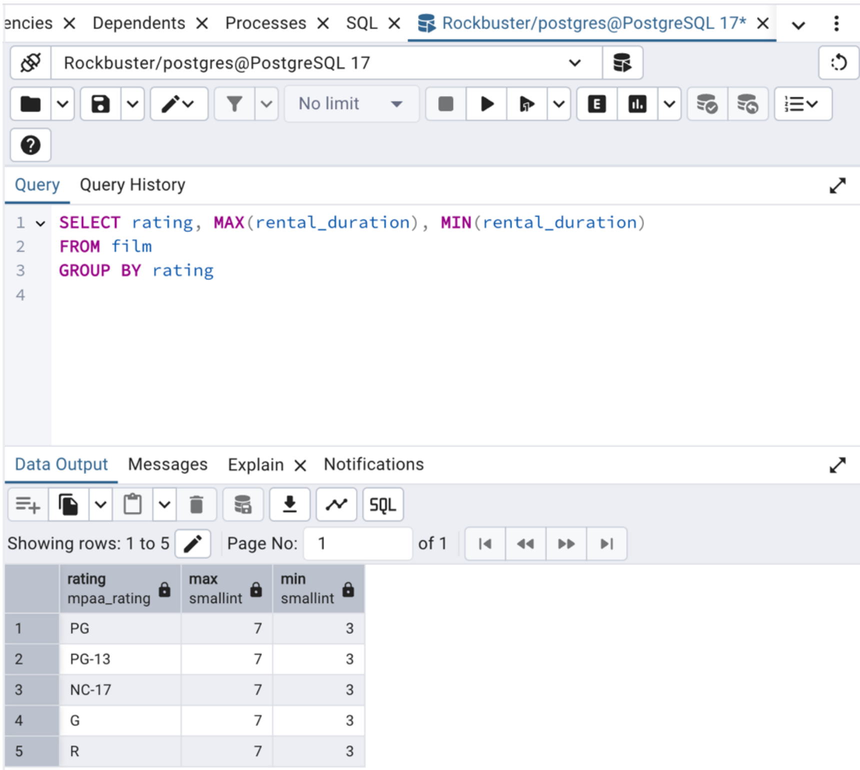
Data Output | Messages | Explain | Notifications

Showing rows: 1 to 5 | Page No: 1 of 1

	rating mpaa_rating	avg numeric
1	PG	3.0518556701030928
2	PG-13	3.0348430493273543
3	NC-17	2.9709523809523810
4	G	2.8888764044943820
5	R	2.9387179487179487

Total rows: 5 | Query complete 00:00:00.097

3a



3b.

4a.

This is the process of ETL- for the Data Engineer

-Extract, which involves collecting the data from multiple data source.

-Transform, the extracted data and convert it into another format.

-Load, where the transformed data is inserted into the new database which is Rockbuster's data base.

4b. There could be lots of problems if the data hasn't gone through the ETL process like missing or incomplete data, data duplications and could be a total waste of time.